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Sedimentology of Late Holocene fluvial levee and point-bar deposits from the Cambodian tract of the Mekong River



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Abstract: The study of channel margin deposits is crucial to better understand fluvial systems and has significant social and economic implications. This paper describes sedimentological features of Late Holocene levee and point-bar deposits from five excellent outcrop exposures along the Cambodian tract of the Mekong River. Point-bar deposits show a typical fining-upward trend with low-angle inclined bedding and consist of gravelly sand, sand and mud beds. Levee deposits show a typical coarsening-upward or no clear vertical grain-size trend and consist of mud and very fine sand beds. Channel margin deposits formed in the outer river bend show a simple levee element, whereas inner bend deposits show more complex architecture with combinations of point-bar and levee elements. Water-level fluctuations play a fundamental role in the construction of point-bar and levee architecture; the resulting deposits show evidence of strong river currents with rapid sedimentation alternating with evidence of subaerial exposure with no sedimentation. This combination of sedimentary features can also form in other environments, but is distinctive of channel margin deposits forming in climates with a pronounced seasonality, including large and perennial rivers in tropical savannah climate areas.

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Fluvial systems have been intensely studied from both modern and ancient examples, leading to the creation of relatively well-developed facies models (e.g. Bridge 2009; Miall 2013). Most studies have focused on channel (e.g. Corbeanu *et al.* 2001; Gibling 2006), floodplain (e.g. Kraus & Gwinn 1997; Varela *et al.* 2012; Burns *et al.* 2017) and point-bar (e.g. Thomas *et al.* 1987; Smith *et al.* 2009; Wu *et al.* 2015; Ghinassi *et al.* 2016) deposits, whereas levee deposits are less studied. Nevertheless, levees play a fundamental role in regulating flooding of low-lying areas and sediment exchange between the river and the floodplain (Brierley *et al.* 1997). Furthermore, because levees consist of elevated areas, they have been fundamental for human settlement, road infrastructure and local economy (e.g. Funabiki *et al.* 2012). Point bars are essential features for the evolution of river bends and their deposits constitute important reservoirs in the geological rock record (e.g. Late Permian Toolachee Formation, Cooper Basin, Australia, Martin & Hamilton 1981; Lower Cretaceous McMurray Formation, Athabasca Oil Sands, Alberta, Canada, Strobl *et al.* 1997). Vertical and lateral facies variability in point-bar deposits has a primary control on reservoir quality and connectivity (e.g. mud layers may act as baffles and barriers). The study of levee and point-bar deposits is crucial to better understand fluvial systems, their stratigraphic evolution and sediment dispersal patterns in their alluvial and deltaic plains. This will in turn help in refining depositional and reservoir models and mitigating flooding risk, and thus has significant economic and social implications.

This paper describes Late Holocene levee and point-bar deposits from five excellent outcrop exposures along the Cambodian tract of the Mekong River (Fig. 1). The main goals of this study are to (1) describe sedimentological features of levee and point-bar deposits, (2) distinguish different types of channel margin deposits and (3) interpret depositional processes and their temporal changes.

Regional background

The Mekong River has a length of *c.* 4350 km and a drainage area of *c.* 810 000 km^2. It originates in the mountainous area of the Tibetan Plateau, in the 'Three Rivers Source Area', together with the Changjiang (Yangtze) and the Huang He (Yellow) rivers (Fig. 1). From its origin, it flows through China, Myanmar, Laos, Thailand, Cambodia and Vietnam and debouches into the South China Sea. From its apex, located near Phnom Penh (Cambodia), the Mekong River forms a mixed-energy delta (i.e. controlled by tide, wave and fluvial processes; Tamura *et al.* 2012; Gugliotta *et al.* 2017). The backwater zone, whose landward limit is taken to be where sea-level intersects the riverbed profile (Paola & Mohrig 1996), extends to Kratie (Cambodia), *c.* 560 km upstream of the river mouth (Gugliotta *et al.* 2017). In the study area, the deposits range in age from 3.5 ka to present day (Oketani & Haruyama 2011) and formed during the Middle to Late Holocene progradation of the Mekong system (Nguyen *et al.* 2000, 2010; Tamura *et al.* 2009; Hanebuth *et al.* 2012), which followed an Early to Middle Holocene transgression that caused the migration of the coastline into Cambodia (Tamura *et al.* 2007; Tjallingii *et al.* 2014).

The study area is characterized by a tropical savannah climate in the Köppen climate classification. Temperature is relatively constant through the year, and seasonal fluctuations mainly consist of changes in wind direction and amount of precipitation driven by the monsoon. Average monthly temperature ranges from 26°C (January) to 31°C (April). The precipitation regime is characterized by a dry season from November to April and a wet season from May

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Table 1. *Water discharge, sediment discharge and water levels of the modern Mekong River*

Annual water discharge	*c.* 550 km^3 a^{-1}
Annual sediment discharge	*c.* 110 million t a^{-1}
Water discharge at Phnom Penh (October)	36700 m^3 s^{-1}
Water discharge at Phnom Penh (May)	2220 m^3 s^{-1}
Water level at Phnom Penh (September 2016)	7.15 m
Water level at Phnom Penh (April 2016)	0.69 m

to October. Water and sediment discharge are *c.* 550 km^3 a^{-1} and *c.* 110 million t a^{-1}, respectively, and vary seasonally (Milliman & Farnsworth 2011; Table 1). At Phnom Penh, the flow varies from 36 700 m^3 s^{-1} in September to 2220 m^3 s^{-1} in April, and the water level fluctuates seasonally up to 10 m (Mekong River Commission 2005; Gugliotta *et al.* 2017). During the dry season, tide-influenced water-level changes are observed up to Phnom Penh and highly diluted brackish water intrudes up to 160 km upstream of the river mouth (Gugliotta *et al.* 2017). At the present day, the study area experiences year-round freshwater conditions and absent to minimal tide-induced water-level fluctuations.

Study area and methods

A field survey was carried out from 29 January to 3 February 2017 during the dry season; detailed stratigraphic sections were measured at five locations from the exposed channel margins along the Cambodian tract of the Mekong River (CA1–CA5; Figs 1 and 2). The sections were measured and described with conventional sedimentological logging techniques and represented at 1:10 scale. Ten bed types were differentiated based on grain size, sorting, colour, sedimentary structures and biogenic features, and were interpreted in terms of depositional processes and environments (for detailed descriptions of each bed type see Table 2). The most upstream section (CA1) is *c.* 57 km downstream of Kampong Cham and *c.* 48 km upstream of Phnom Penh, whereas the most downstream section (CA3) is *c.* 9 km downstream of Phnom Penh and *c.* 47 km upstream of Neak Luong (note: distances were measured at the midpoint of channel width along the longitudinal river axis). The geomorphological features of the tracts of the river where the sections are located were analysed using 2017 satellite images from Google Earth Pro and a digital elevation model with a grid resolution of 30 × 30 m from the Shuttle Radar Topography Mission (SRTM) obtained from the United States Geological Survey (USGS) Earth Explorer data portal (https://earthexplorer.usgs.gov/). All the studied sections have experienced recent bank erosion and as a result they show well-exposed outcrops with thickness varying from 7 m (CA4) to 8.9 m (CA1) that generally decreases downstream (Figs 2 and 3). Section CA2 was examined in two parts: the upper 2 m (CA2b) were examined *c.* 150 m to the SE of the major section (CA2a; Fig. 2b). Each section was logged from the water level on the day of the field survey to the top. During the field-survey period, the water level was *c.* 1 m above the minimum water level of the 2015–2016 dry season from April 2016 and *c.* 5–7 m below the maximum of the 2016 wet season from September 2016 (Fig. 4). Maximum and minimum water levels were calculated from the data of the hydrological stations of Kampong Cham, Phnom Penh and Neak Luong taken from the Mekong River Commission data portal (http://portal.mrcmekong.org/index).

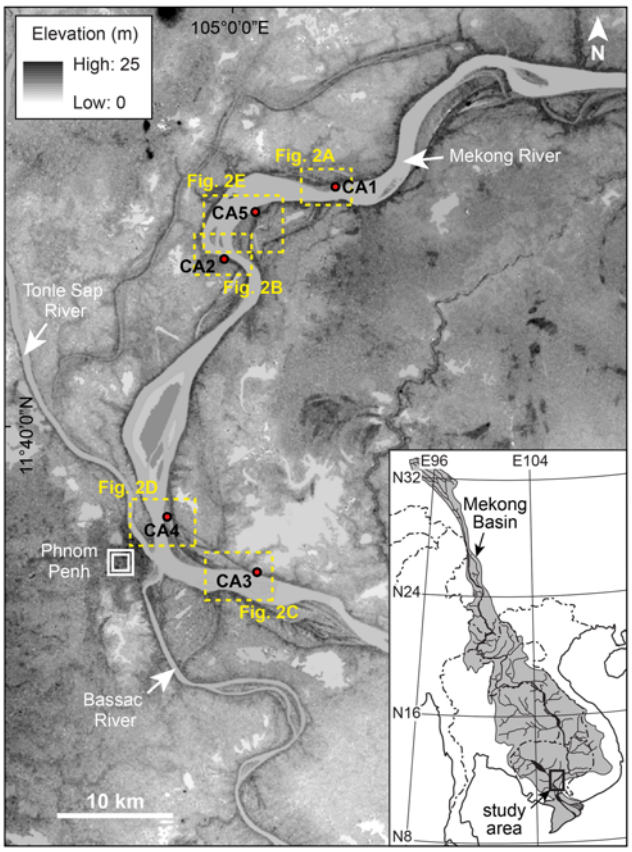


Fig. 1. Digital elevation model of the study area with locations of the five studied sections (CA1–CA5). The grey area in the inset map shows the drainage basin of the Mekong River.

Results

Section CA1

Description

CA1, the most upstream section, is located *c.* 48 km upstream of Phnom Penh, in the upstream part of an active point bar (Fig. 2a). The section consists of 8.7 m thick succession of gravelly sand, sand and mud deposits. It shows an overall fining-upward trend for the sand–gravel component, with granules and pebbles present in only the lower 3.6 m of the section, but mud content generally decreases upward (Fig. 5). The top of the section consists of a 0.15 m thick modern soil. The bedding shows a low-angle inclination toward the channel axis and the downstream direction of the river (i.e. inclined heterolithic stratification; IHS). Bed contacts are usually erosional or sharp throughout the entire section and, in a few cases, show signs of oxidation. In more detail, the lowermost 4 m of the section consists of two 2 m thick fining-upward units, whereas the uppermost 4.5 m shows no clear vertical grain-size trend. The lowest unit consists of alternations of mud beds (B2 and B3) and sand beds (B6, B7 and B9). Sand beds are up to 0.25 m thick, structureless or cross stratified, and show small mud clasts. The upper fining-upward unit consists of alternations of gravelly sand beds (B10) and mud beds (B2 and B4), with one bed of sand with climbing-ripple lamination (B5). The B10 beds are up to 0.35 m thick and shows tabular-cross stratification and mud clasts; the gravel component consists of granules and pebbles up to 1 cm in diameter. Inverse graded mud beds (B4) are up to 0.35 m thick, show parallel lamination, sand–mud interlamination and ripple sand lenses (i.e. lenticular bedding), and pass upward to B5 or B10 across a gradual contact. The uppermost 4.5 m of the section consists of very fine sand beds (B5 and B7) with few intercalated mud layers (B3 and B4; Fig. 6g). B5 shows climbing-ripple (Fig. 6f) and parallel lamination, soft-sediment deformation and local bioturbation, and can also show normal or inverse grading (Fig. 5). B7 is structureless and contains abundant small (up to 2 cm in diameter) and rounded mud clasts. Mud beds are typically structureless, or

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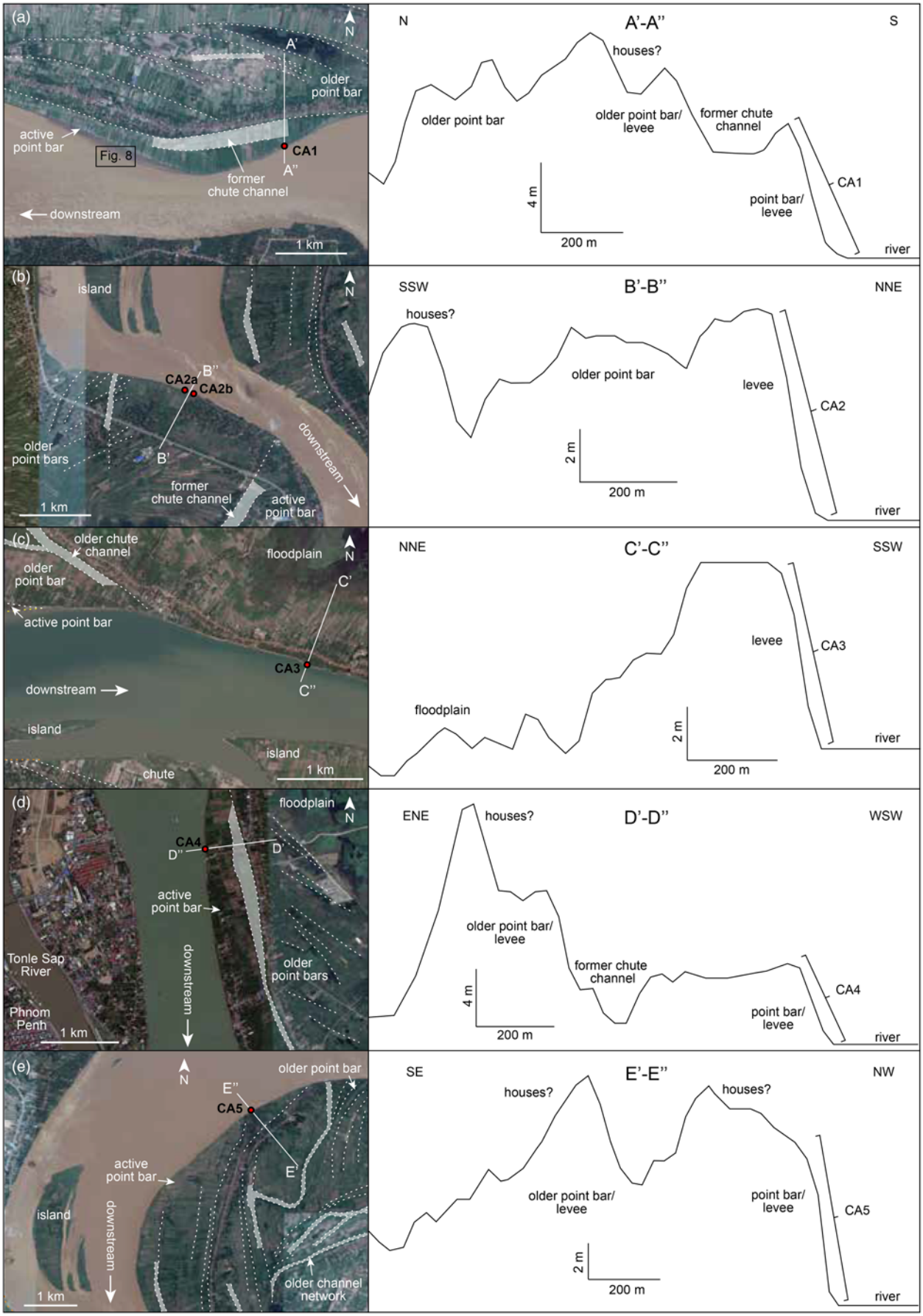


Fig. 2. River bend configuration and elevation profile of the channel margin at each of the five studied locations. The elevation data are derived from the digital elevation model of Figure 1. The satellite images are from Google Earth Pro. (See Fig. 1 for location of each section.)

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Table 2. *Description of the 10 bed types identified in this study*

Bed type	Grain size	Contacts	Sedimentary structures	Other features	Processes	Environment
B1: grey mottled mud	Clay to fine silt	Gradual or poorly defined (lower and upper)	Structureless, local parallel lamination	Slickenside, pedogenic mottling, some plant matter	Flooding and subaerial exposure	Floodplain
B2: grey to brown mud	Clay to fine silt	Sharp or erosional and can show signs of oxidation (lower and upper)	Structureless, parallel lamination	Local bioturbation, some plant matter	Waning flooding phase	Point bar
B3: organic-rich red to purple mud	Clay to silt with some very fine sand	Gradual and poorly defined or sharp (lower and upper)	Structureless, parallel lamination, sand–mud interlamination, lenticular bedding	Pedogenic mottling, abundant plant matter, root traces	Flooding and subaerial exposure	Levee and upper point bar
B4: inverse graded mud	Clay to silt with some very fine sand	Sharp or erosional with signs of oxidation (lower) and either gradual or sharp/ erosional (upper)	Inverse grading, structureless, parallel lamination, sand–mud interlamination, lenticular bedding	–	Waxing flooding phase	Levee and point bar
B5: sand with climbing-ripple and/or parallel lamination	(Muddy) very fine sand, commonly well sorted	Sharp or erosional (lower and upper), but lower contact can be gradational when overlying B4	Climbing-ripple lamination, parallel lamination, local soft-sediment deformation, normal and inverse grading	Some small mud clasts, root traces, local bioturbation	Flooding and rapid deposition	Levee and upper point bar
B6: structureless sand	Very fine to fine sand, commonly well sorted	Sharp or erosional (lower) and poorly defined (upper)	Structureless	–	Flooding	Levee and upper point bar
B7: sand with small rounded mud clasts	(Muddy) very fine to fine sand	Sharp or erosional (lower and upper)	Structureless	Abundant small rounded mud clasts	Flooding and local erosion	Levee and upper point bar
B8: sand with large tabular mud clasts	Fine to medium sand	Sharp or erosional (lower and upper)	Structureless	Large tabular mud clasts	Reworking of desiccation mud cracks	Point bar
B9: sand with cross stratification	Fine to medium sand	Sharp or erosional (lower and upper)	Trough cross stratification, planar cross stratification, structureless	Some mud clasts	River bedload	Lower point bar
B10: gravelly sand	Medium to coarse sand with granules and pebbles	Sharp or erosional (lower and upper)	Trough cross stratification, planar cross stratification, structureless	Some mud clasts	River bedload	Lower point bar

with parallel lamination or sand–mud interlamination, and show root traces, plant matter and oxidized colours (i.e. red to purple; Fig. 6c).

Interpretation

The grain-size range and sedimentary structures suggest that these sediments were deposited either as subaqueous dunes (B9 and B10) or during flooding events (B2, B3, B4, B5, B6 and B7). B4 record increasing current strength and rising water level. Climbing-ripple lamination and soft-sediment deformation in B5 suggest rapid sedimentation. The small mud clasts of B7 were eroded from mud-rich parts of the channel margin during the flooding processes. The presence of root traces, oxidized colours and bioturbation suggests periods of subaerial exposure and/or non-deposition. B2 formed in marginal channel areas with lower elevation and that were less

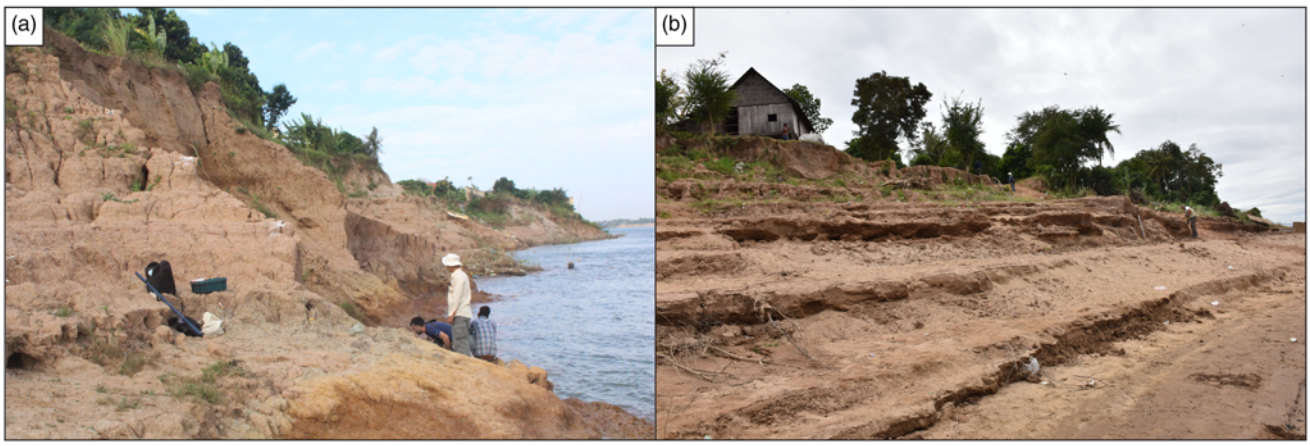


Fig. 3. (**a**) Section CA2a showing grey to purple mud deposits. (**b**) Section CA5 showing low-angle inclined sand and mud alternations.

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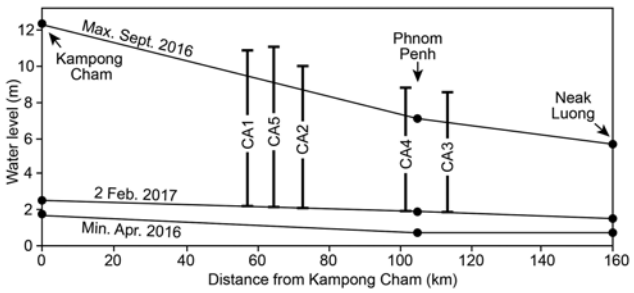


Fig. 4. Reconstructions of water levels from April 2016 (minimum of the 2015 – 2016 dry season), September 2016 (maximum of the 2016 wet season) and February 2017 (surveying period) with locations and vertical distribution of the five studied sections.

well-drained compared with B3. Based on its present-day position in relation to the river bend, vertical trends, bed types and their association, grain size and sedimentary structures, this section is interpreted to represent a combination of channel thalweg and point-bar (lower part) to levee (upper part) environments formed in an inner river bend (Fig. 7). The broad fining-upward trend and the low-angle inclination are the result of the channel migration and the consequent accretion of the point bar over thalweg deposits. The subdivision of the lower part of the section into two units may be due to a rearrangement of the channel owing to minor changes in flow direction or other local modifications. The prevalence of sand beds and the presence of abundant mud clasts in the upper part of the section suggest that the overlying levee deposits formed near the river channel.

Section CA2

Description

CA2 is located *c.* 32 km upstream of Phnom Penh, in a cut bank area upstream of an active point bar (Fig. 2b). The section consists of an 8 m thick succession of mud and sand deposits and shows an overall coarsening-upward trend from clay to very fine sand (Fig. 5). The top of the section consists of a 0.5 m thick artificial sand derived from dredging of the river bed. The bedding is nearly horizontal and bed contacts are gradual or poorly defined in mud beds, but sharp or erosional at the base of the sand beds (Fig. 3a). The lowermost 2.4 m of the section consists of alternations of grey mottled mud (B1) and

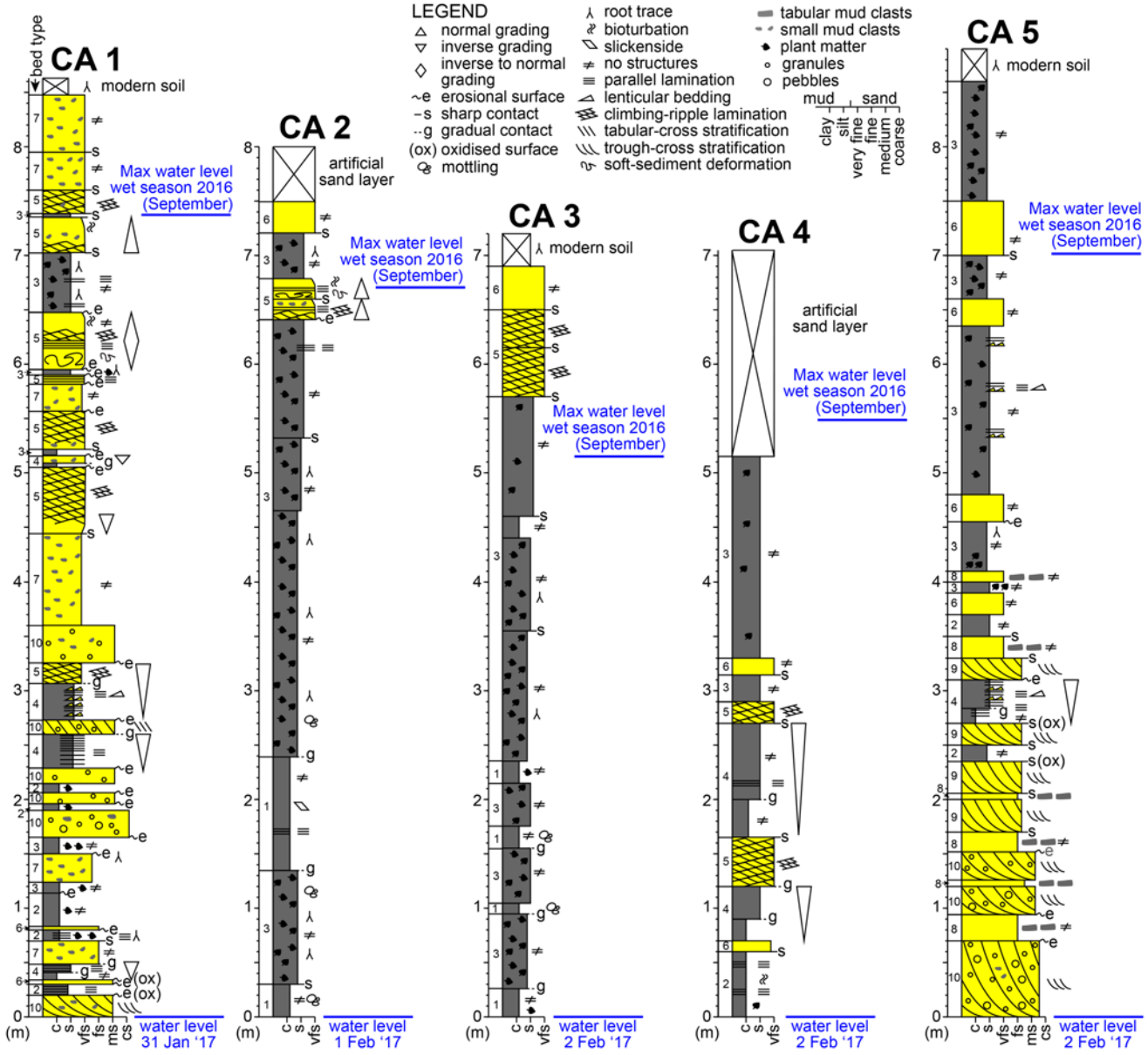


Fig. 5. Sedimentary logs of the five studied sections with reported water level during the survey and maximum water level of the 2016 wet season.

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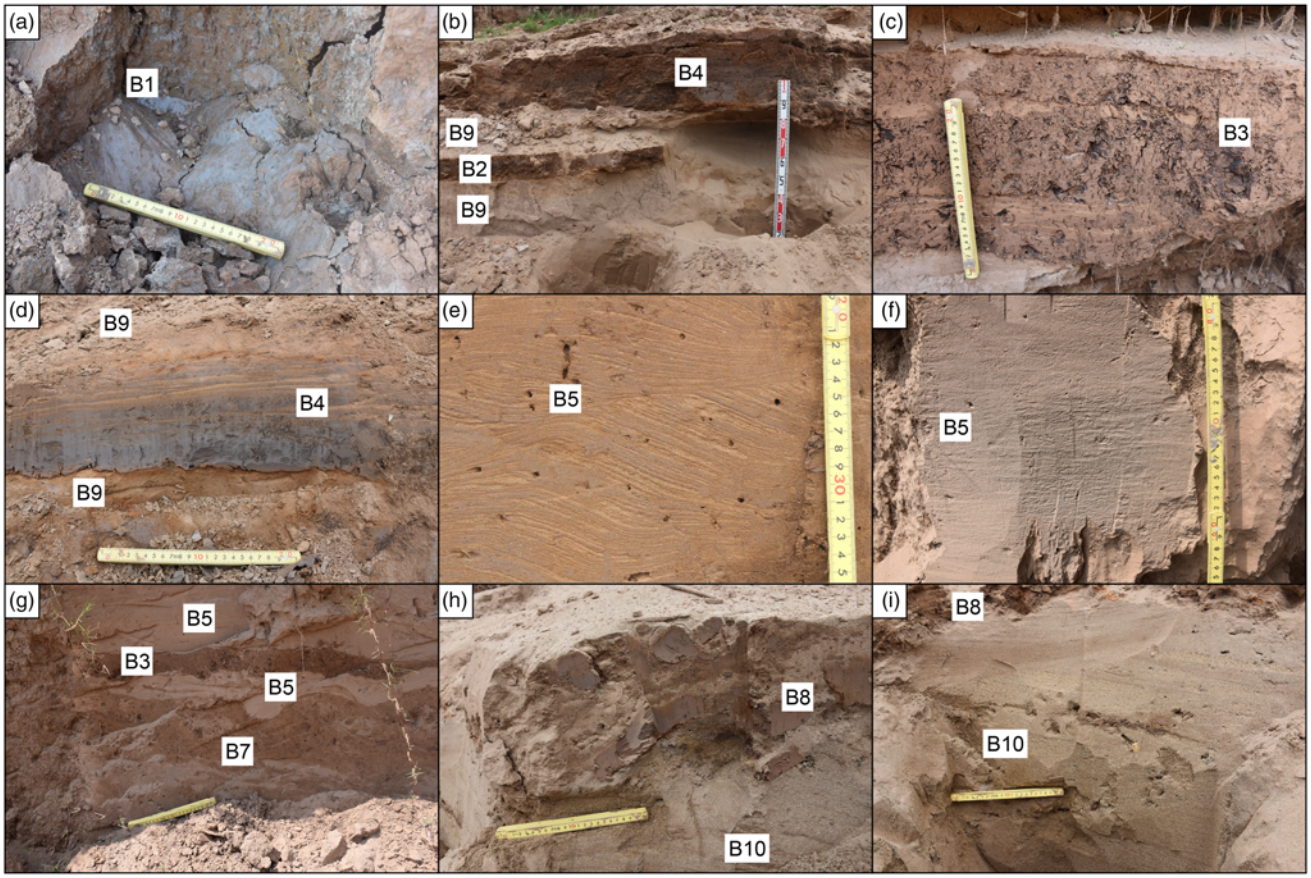


Fig. 6. Representative photographs of the bed types from the studied section. (**a**) Mottled mud and slickenside (B1). (**b**) Alternations of trough cross-stratified sand (B9) and brown mud (B2) or inverse graded mud (B4). (**c**) Red to purple mud with plant matter, root traces and local sand laminae (B3). (**d**) Detail of inverse graded mud bed (B4) changing upward from structureless to laminated and sand–mud interlaminated, overlying and overlain by trough cross-stratified sand (B9). (**e**, **f**) Sand with climbing-ripple lamination (B5). (**g**) Alternations of red to purple mud (B3) or sand with mud clasts (B7) and sand with climbing-ripple lamination (B5). (**h**) Structureless sand with large tabular mud clasts (B8), overlying trough cross-stratified gravelly sand (B10). (**i**) Detail of trough cross-stratified gravelly sand (B10). It should be noted that the scale of each photograph (except (**b**)) shows numbered tags each centimetre. In (**b**), the scale bar shows numbered tags each decimetre.

organic-rich red to purple mud (B3). The part of the section between 2.4 and 6.4 m is characterized by B3. The uppermost 1 m show alternations of B3 and very fine sand beds (B5 and B6). Mud beds (B1 and B3) are typically structureless, with local parallel lamination or sand–mud interlamination, and show diffuse root traces, plant matter, pedogenic mottling and oxidized colours. B1 also shows slickensides (Fig. 6a). Sand beds range in thickness from 0.2 to 0.3 m, can show normal grading and either display climbing-ripple lamination, parallel lamination, local bioturbation and soft-sediment deformation (B5), or are structureless (B6).

Interpretation

The grain-size range and sedimentary structures suggest that these sediments were deposited rapidly during flooding events. Diffuse pedogenic features suggest relatively long periods of subaerial exposure. B1 represents floodplain areas that are more distal in relation to the river, and have lower elevation and are less well-drained compared with B3. B5 and B6 at the top of the section represent stronger flooding events and/or increasing proximity to the river. Based on the coarsening-upward trend, bed types and their association, grain-size range and sedimentary structures, this section is interpreted to represent floodplain (lower part) to levee (upper part) environments formed in an outer river bend (Fig. 7). The clear coarsening-upward trend is due to the migration of the river toward the bank and the consequent migration of the levee over floodplain deposits. The prevalence of mud sediments with few intercalated sand beds and the presence of floodplain deposits suggest that these deposits formed in a rather distal location in relation to the river channel.

Section CA3

Description

CA3, the most downstream section, is located *c.* 9 km downstream of Phnom Penh, in a cut bank area downstream of an active point bar (Fig. 2c). The section consists of 7.2 m thick succession of mud and sand deposits and shows an overall coarsening-upward trend from clay to very fine sand (Fig. 5). The top of the section consists of a 0.3 m thick modern soil. The bedding is nearly horizontal and bed contacts are gradual or poorly defined in mud beds, but sharp or erosional at the base of the sand beds. The lowermost 5.7 m of the section consists of alternations of grey mottled mud (B1) and organic-rich red to purple mud (B3), whereas the uppermost 1.2 m of the section is characterized by very fine sand beds (B5 and B6). Mud beds are structureless and show diffuse root traces, plant matter, pedogenic mottling and oxidized colours. Sand beds are from 0.35 to 0.45 m thick and either show climbing-ripple lamination (B5) or are structureless (B6).

Interpretation

As for CA2, the grain-size range, sedimentary structures and post-depositional features suggest that these sediments were deposited rapidly during flooding events and were subjected to relatively long periods of subaerial exposure. Based on the coarsening-upward

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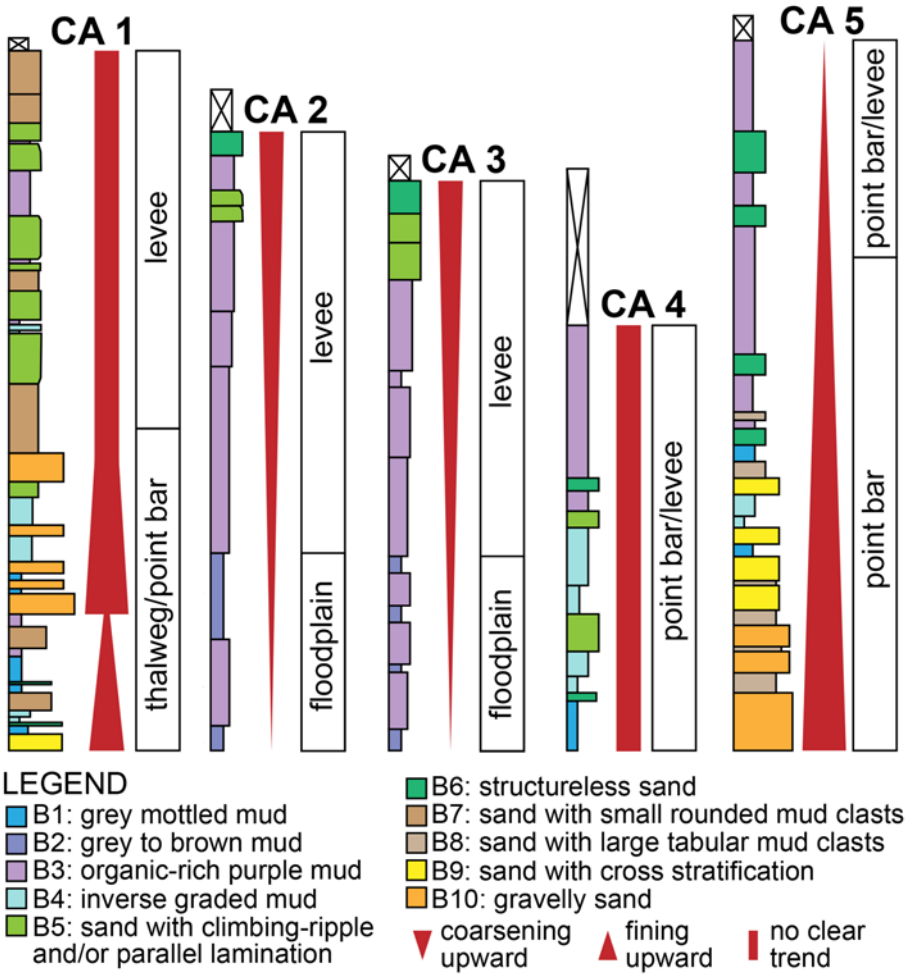


Fig. 7. Bed type, trend and environment interpretations of the five studied sections. Grain size, vertical scale and sedimentary features of the sections are given in Figure 5.

trend, bed types and their association, grain-size range and sedimentary structures, this section is interpreted to represent floodplain (lower part) to levee (upper part) environments formed in an outer river bend and in a rather distal location in relation to the river channel (Fig. 7).

Section CA4

Description

CA4 is located *c.* 3 km upstream of Phnom Penh, in the upstream part of an active point bar (Fig. 2d). The section consists of 7 m thick succession of very fine sand and mud deposits and shows no clear vertical grain-size trend (Fig. 5). The top of the section consists of a 2 m thick artificial sand bed derived from river dredging or other human activities. Bed contacts are sharp in the lower part of the section, but less well defined in the upper part. The lowermost 3.3 m of the section shows alternations of mud beds (B2 and B4) and sand beds (B5 and B6). Mud beds range in thickness from 0.5 to 1 m, and are structureless and bioturbated or with parallel lamination. Sand beds range in thickness from 0.1 to 0.45 m and are structureless (B6) or show climbing-ripple lamination (B5; Fig. 6e). The part of the section between 3.3 and 5.15 m shows organic-rich red to purple mud (B3) with an intercalated structureless 0.15 m thick very fine sand bed (B6).

Interpretation

The grain-size range and sedimentary structures suggest that most of these sediments were deposited rapidly during flooding events. Pedogenic features in the upper part of the section suggest subaerial exposure. The climbing-ripple lamination of B5 and the thick beds of B4 suggest rapid deposition with beds up to 1 m thick deposited in a single event. Based on its present-day position in relation to the river bend, vertical trends, bed types and their association, grain size and sedimentary structures, this section is interpreted to represent upper point-bar to levee environments formed in an inner river bend (Fig. 7).

Section CA5

Description

CA5 is located *c.* 40 km upstream of Phnom Penh, in the upstream part of an active point bar (Fig. 2e). The section consists of an 8.9 m thick succession of gravelly sand, sand and mud deposits and shows a typical fining-upward trend, with granules and pebbles present only in the lower part and mud content increasing upward (Fig. 5). The top of the section consists of a 0.3 m thick modern soil. The bedding shows a low-angle inclination toward the channel axis and downstream direction of the river (IHS; Fig. 3b). Bed contacts are erosional or sharp in the lowermost 3.5 m of the section, but less well defined in the uppermost 4 m. The lowermost 2.3 m of the section consists of alternations of trough cross-stratified gravelly sand or sand (B9 and B10) and structureless sand with large tabular mud clasts (B8; Fig. 6h and i). The gravel component consists of granules and pebbles up to a few centimetres in diameter. The tabular mud clasts show relatively angular shape and outline and their longest axis can be several decimetres long. Bed thickness ranges from 0.25 to 0.7 m for B9 and B10 and from 0.05 to 0.25 m for B8. The part of the section between 2.3 and 4.8 m shows various types of mud beds (B2, B3 and B4) and sand beds (B6, B8 and B9; Fig. 6b and d). Mud beds consist of brown to grey mud (B2), organic-rich red to purple mud (B3) and inverse graded mud (B4; Fig. 6d), and range in

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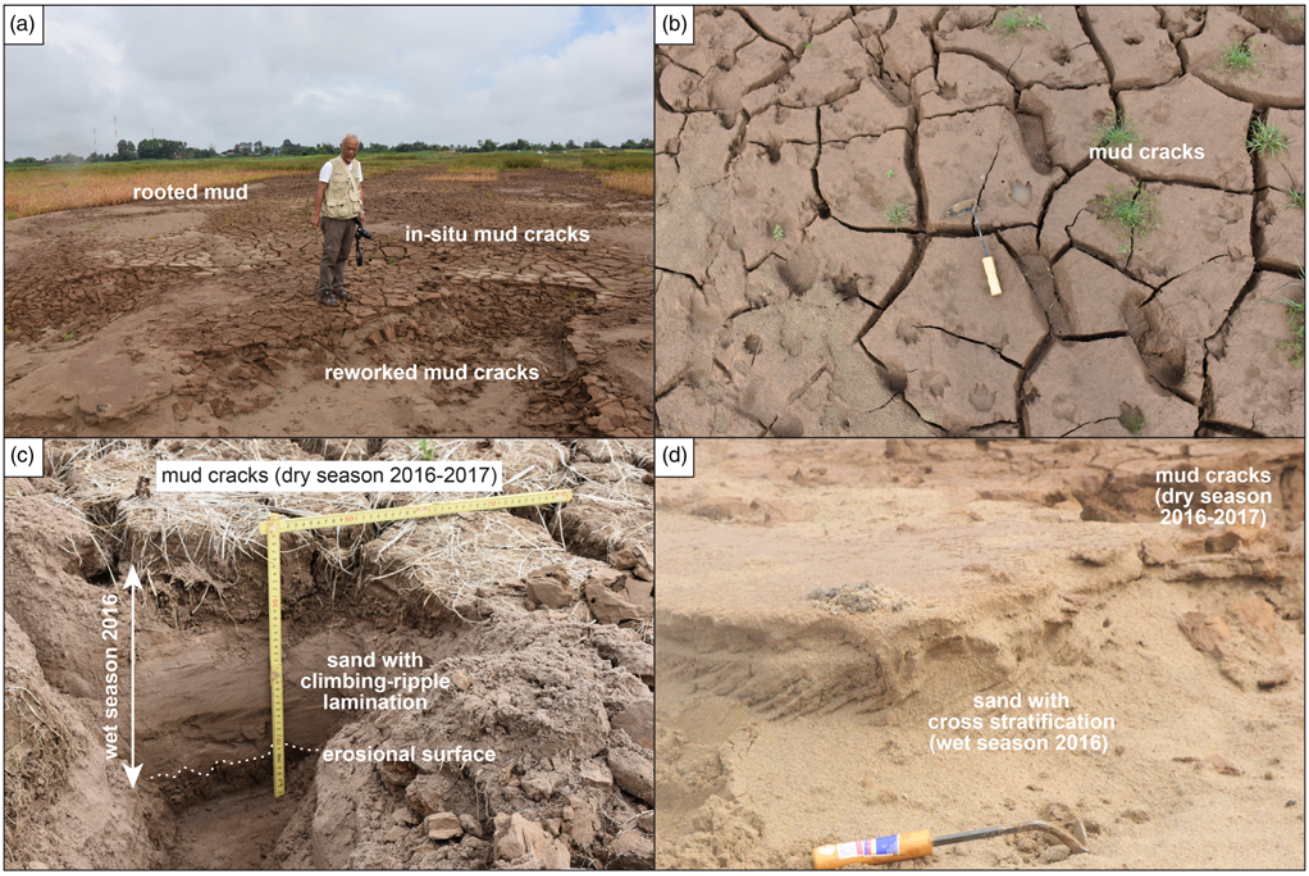


Fig. 8. Photographs from an active point bar observed during the survey period (dry season) a few kilometres to the west of section CA1 (see location in Fig. 2a). (**a**) Extensive mud-crack area. (**b**) Detail of mud cracks. (**c**) Detail of seasonal bedding with climbing-ripple lamination (wet season) and mud cracks (dry season). (**d**) Detail of seasonal bedding with cross stratification (wet season) and mud cracks (dry season). Object for scale in (**b**) and (**d**) is 25 cm long.

thickness from 0.15 to 0.45 m. They are relatively continuous laterally (i.e. tens of metres), but they show lateral changes in thickness and pinch-out terminations (Fig. 6b). Mud beds are structureless or show parallel lamination, sand–mud interlamination and lenticular bedding. Sand beds vary from structureless (B6), with large mud clasts (B8), to trough cross stratified (B9) and range in thickness from 0.1 to 0.25 m. The uppermost 2.8 m of the section shows organic-rich red to purple mud (B3) and 0.25–0.5 m thick structureless very fine sand beds (B6).

Interpretation

The grain-size range and sedimentary structures suggest that these sediments were deposited either as subaqueous dunes (B9 and B10) or during flooding events (B2, B3, B4 and B6). B8 formed because of reworking of desiccation mud cracks and underlying sand beds. The lateral changes in thickness and pinch-out terminations of mud beds are because they tend to fill dune troughs. Based on its present-day position in relation to the river bend, the clear fining-upward trend, bed types and their association, grain size and sedimentary structures, this section is interpreted to represent a point-bar environment formed in an inner river bend; however, the upper part with anomalously thick beds of B6 may represent an overlying mixed point-bar levee environment (Fig. 7).

Discussion

Levee and point-bar sedimentary features and processes

Point-bar deposits show a typical fining-upward trend and low-angle inclined bedding (IHS). Thalweg and lower point-bar deposits consist of alternations of cross-stratified sand (B9) or gravelly sand (B10) and sand with tabular mud clasts (B8) or mud layers (B2 and B4). Upper point-bar deposits consist of mud (B2 and B3) with a few sand beds (B5 and B6). Observations of active point bars within the study area during the surveying period (dry season; Fig. 8) evidenced the presence of extensive mud-cracked areas up to several kilometres wide and sedimentary structures corresponding to the interpreted point-bar deposits (i.e. trough cross stratification, climbing-ripple lamination, mud cracks and mud clasts). Levee deposits show a typical coarsening-upward trend and mud content decreasing upward or no clear vertical grain-size trend. The levee deposits formed near the channel consist of sand deposits (B5, B6 and B7) with few mud intercalations (B3 and B4) and abundant mud clasts and climbing-ripple lamination, whereas the levee deposits formed away from the channel consist of dominant mud deposits (B1 and B3) with a few sand beds (B5 and B6) and more diffuse pedogenic features.

Channel margins are substantially different in the outer and inner sides of river bends (Fig. 9). Outer bends show a simple levee element (sections CA2 and CA3), with a well-defined coarsening-upward trend owing to the channel migration toward the bank and the consequent accretion of the levee over floodplain deposits (Fig. 9). As a result of the erosional conditions, the part of the levee forming near the channel has low preservation potential, whereas the distal part of the levee and the floodplain deposits are more likely to be preserved. Inner bends show more complex architecture with a combination of point-bar and levee elements (sections CA1, CA4 and CA5). Point-bar deposits show a typical fining-upward trend, whereas the overlying levee show no clear vertical grain-size trend. The fining-upward trend is due to the channel migration away

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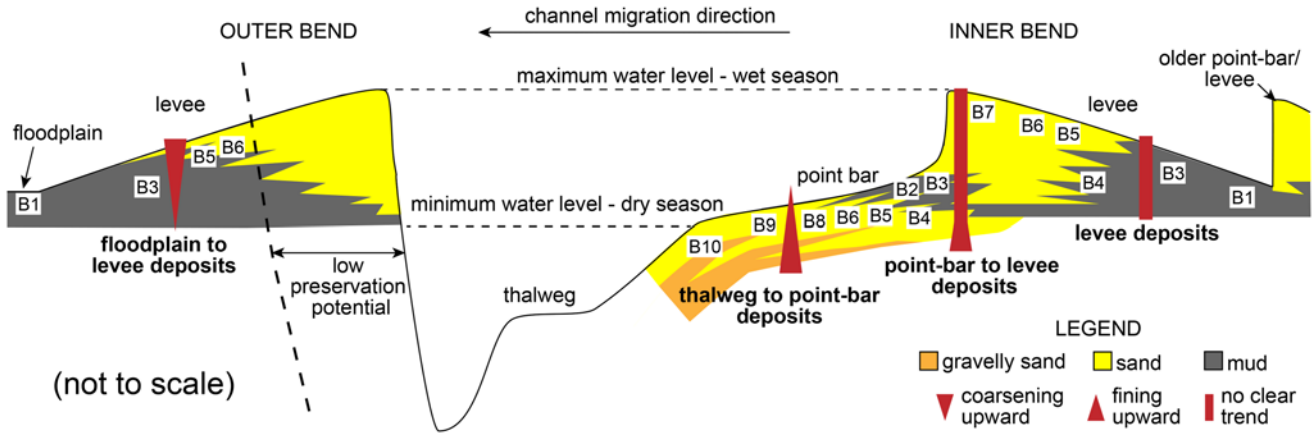


Fig. 9. Conceptual depositional model for the inner and outer river bends, showing the distribution of bed types.

from the bank and the consequent accretion of the point-bar over thalweg deposits (Fig. 9). In the inner bend, more than one levee can be preserved and the proximal part of the levee has a higher preservation potential compared with the levees forming in the outer bend.

In levee and upper point-bar deposits, the grain size restricted between clay and very fine sand, the presence of climbing-ripple lamination and up to 1 m thick mud beds indicate that most of these sediments were deposited rapidly as a combination of bedload and suspended load during decelerating flooding events (see Ashley *et al.* 1982). In contrast, inverse-graded mud beds indicate waxing currents and rising water levels and resemble similar features reported from other river banks (see Iseya 1989; Iseya & Ikeda 1989). Sedimentation rate is generally high, but probably decreases with distance from river channel (see also Goodbred & Kuehl 1998; Walling & He 1998). Small and rounded mud clasts of B7 formed because of erosion of mud from the channel banks during flooding events. Because of their short life-span when transported for long distances (Smith 1972) this type of mud clasts is found in channel margin deposits forming near the river and its abundance decreases with distance away from it (i.e. toward the floodplain or the channel axis). In thalweg and lower point-bar deposits, the presence of gravels, coarser sand and cross stratification indicate that these sediments were deposited as river bedload (see Claude *et al.* 2012). The tabular mud clasts of B8 formed because of reworking of desiccation mud cracks. The tabular shape and large size are because these mud clasts are derived from mud beds up to several decimetres thick that were subjected to relatively long periods (i.e. several months) of subaerial exposure. Root traces, pedogenic features, oxidized colours, bioturbation and mud cracks indicate frequent and relatively long periods of subaerial exposure and/or non-deposition. Sharp and erosional contacts between beds originated either during the subaerial exposure or as a result of strong erosional currents, or by a combination of the two processes.

Temporal changes of depositional processes

To better understand how these deposits formed, it is fundamental to consider the temporal changes of depositional processes and their role in the construction of levee and point-bar architecture. Five phases were identified (Fig. 10), although they are rarely represented altogether in the same cycle. In levees, deposition of mud (B4) starts in the early flooding phase, with waxing currents and rising water levels, and more specifically when the water level is high enough to spill over channel levees. With increasing strength, the flow may become non-depositional and produce a subaqueous erosional surface. In the portion of the levee near the river, erosion creates abundant mud clasts of B7. As the flows starts to wane, sand (B5, B6 and B7) is deposited over the levee, whereas mud (B1 and B3) reaches the distal part of the levee and the floodplain. In the late flooding phase, with waning currents and falling water level, mud (B3) is deposited above the levee. Finally, during the inter-flooding phase, the levee is subaerially exposed favouring oxidation, pedogenic processes, and the development of plants and of a subaerial erosional surface. Mud sediments are almost entirely modified by post-depositional processes. In point bars, deposition of mud (B4) starts in the early flooding phase, with waxing currents and rising water levels. With increasing strength, the motion of sand and gravel starts, but the flow is non-depositional and produces a subaqueous erosional surface. As the flows starts to wane, sand and gravelly sand are deposited in subaqueous dunes (B9 and B10) in thalweg and lower point-bar areas, whereas a bed made of sand (B5 and B6) passing to mud (B2 and B3) is deposited in the upper part of the point bar. In the late flooding phase, with waning currents and falling water levels, mud (B2 and B3) is deposited across most of the point-bar area. Finally, during the inter-flooding phase, a large part of the point bar is subaerially exposed, favouring oxidation, pedogenic processes and the development of mud cracks, plants and of a subaerial erosional surface. Mud cracks are later reworked to form B8 at the beginning of the following cycle.

In levees, although flooding events always occur during the wet season, these are not regular. They may not occur for one or more seasons if the maximum water level is lower than the elevation of the levee, or they can occur several times within the same season if there are local fluctuations of the water level owing to changes in rainfall intensity. Levee deposits show non-regular and stochastic alternations of beds forming under different conditions, rather than a clear seasonal signature. In point bars, each flooding and inter-flooding phase corresponds to the wet and dry season, respectively. Point-bar deposits show regular and predictable alternations of beds forming under different conditions. These alternations represent seasonal cycles in sedimentation and share affinities with interpreted seasonal beddings from other modern and ancient systems (Sisulak & Dashtgard 2012; Dalrymple *et al.* 2015; Gugliotta *et al.* 2016; Jablonski & Dalrymple 2016). Whether the alternations of different conditions are driven by seasonal changes or irregular flooding events, water-level fluctuations play a fundamental role in the construction of point-bar and levee architecture. In the Cambodian tract of the modern Mekong River, these water-level fluctuations are up to 10 m and are driven by the monsoon of a tropical savannah climate. In other areas, they can be driven by other types of precipitation regimes or snowmelt (Milliman & Farnsworth 2011). The resulting deposits show evidence of strong river currents (e.g. gravel bed load) and rapid deposition (e.g. climbing-ripple lamination) alternating with evidence of subaerial exposure and non-deposition (e.g. root traces). This combination of sedimentary

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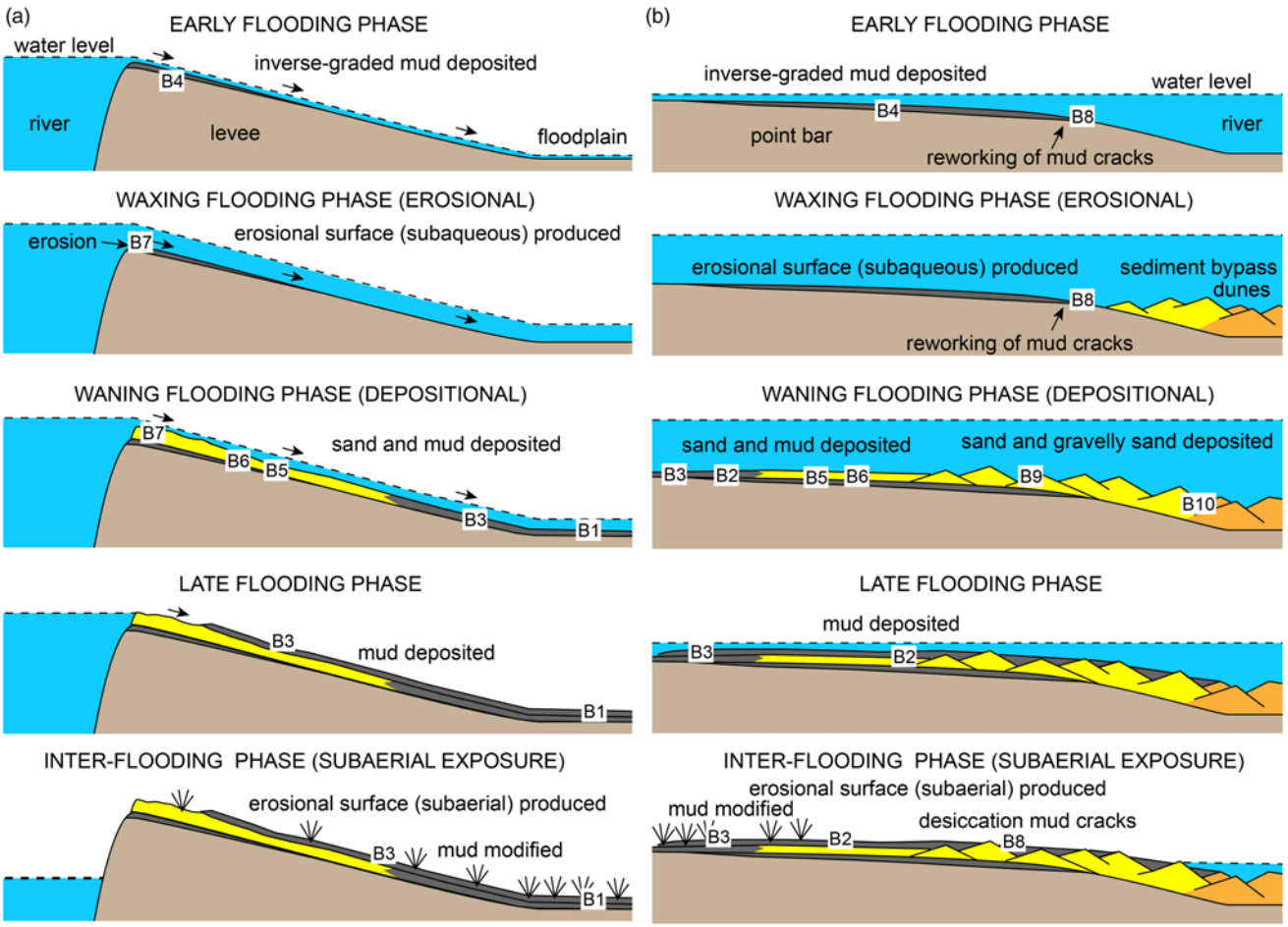


Fig. 10. Conceptual depositional model showing sedimentary processes in levee (**a**) and point-bar (**b**) areas under different conditions through time.

features can also form in other environments, but is distinctive of channel margin deposits forming in climates with a pronounced seasonality, including large and perennial rivers in tropical savannah climate areas.

Conclusions

(1) This study describes in detail the sedimentary features of levee and point-bar deposits along the Cambodian tract of the Mekong River. Point-bar deposits show a typical fining-upward trend and low-angle inclined bedding. Thalweg and lower point-bar deposits consist of alternations of cross-stratified sand or gravelly sand and sand with tabular mud clasts or mud layers. The upper point-bar deposits consist of mud with a few sand beds. Levee deposits consist of mud and very fine sand and show a typical coarsening-upward or no clear vertical grain-size trend and mud content decreasing upward.

(2) Channel margins are substantially different in outer and inner river bends. Outer bends show a simple levee element, with a well-defined coarsening-upward trend owing to the channel migration toward the bank and the consequent accretion of the levee over floodplain deposits. Inner bends show more complex architecture with a combination of point-bar and levee elements. Point-bar deposits show a typical fining-upward trend, whereas the overlying levee shows no clear vertical grain-size trend. The fining-upward trend is due to the channel migration away from the bank and the consequent accretion of the point bar over thalweg deposits.

(3) In levee and upper point-bar deposits, sediments were deposited rapidly during flooding events. In thalweg and lower point-bar deposits, sand and gravelly sand sediments were deposited as subaqueous dunes. Root traces, pedogenic features, oxidized colours, bioturbation and mud cracks indicate frequent and protracted periods of subaerial exposure and/or non-deposition. Sharp and erosional contacts between beds originated either during the subaerial exposure or as a result of strong erosional currents, or by a combination of the two processes.

(4) Both levee and point-bar architecture were constructed by temporally evolving depositional processes. Deposition of mud starts in the early flooding phase, with waxing currents and rising water levels. With the increase of water level and of the flow, after a non-depositional and erosional phase, sand and mud are deposited on the levee and upper point bar, whereas sand and gravelly sand are deposited as subaqueous dunes in thalweg and lower point-bar areas. In the late flooding phase, with waning currents and falling water levels, mud is deposited in a large part of the margin areas. Finally, during the inter-flooding phase, the margin area is subaerially exposed, favouring oxidation, pedogenic processes, and the development of plants and mud-crack areas.

(5) In levees, although flooding events always occur during the wet season, these are not regular; levee deposits show non-regular and stochastic alternations of beds forming under different conditions, rather than a clear seasonal signature. In point bars, each flooding and inter-flooding phase corresponds to the wet and dry seasons, respectively; point-bar deposits show regular and predictable alternations

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of beds forming under different conditions and representing seasonal cycles in sedimentation.

(6) Water-level fluctuations play a fundamental role in the construction of point-bar and levee architecture. In the Cambodian tract of the modern Mekong River, these water-level fluctuations are up to 10 m and are driven by the monsoon of a tropical savannah climate. The resulting deposits show evidence of strong river currents and rapid deposition alternating with evidence of subaerial exposure and non-deposition. This combination of sedimentary features can also form in other environments, but is distinctive of channel margin deposits forming in climates with a pronounced seasonality, including large and perennial rivers in tropical savannah climate areas.

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